Reg. No.: 22BAT1266

Name : AFZAL

VIT

Vellore Institute of Technology

(Deemed to be University under section 3 of UGC Act, 1956)

Continuous Assessment Test II (CAT II) – May 2023

Programme	:B.Tech. Computer Science and Engineering	Semester	:	Winter Semester 2022-23
Course Code	:BCSE102L	Class Nbr(s)	:	CH2022232300780 CH2022232300566 CH2022232300569 CH2022232300567 CH2022232300735 CH2022232300568 CH2022232300782 CH2022232300725
Course Title	:Structured and Object-Oriented Programming			
Faculty(s)	:Dr.A.Menaka Pushpa Dr.Vatchala .S Dr. S. Pavithra Dr.Ilavendhan .A Prof.R.Sujithra Dr. D. Sathian Dr. T. Raja Sree Dr. Christopher Columbus C	Slot	:	C2
Time	:90 Minutes	Max. Marks	:	50

Answer all the Questions

Q. No.	Sub-division	Question Text	Marks
1.		Jack is a sports teacher at St. Patrick's School. He makes games not only to make the student fit but So smart. So, he lined up all the N number classes of students in his class. At each position, he has fixed a board with the Integer number printed on it. Each number is unique and in exactly the range of N. Let us say there are 10 students, then the boards will be printed with numbers from 1 to 10 in a random order given by the sequence A[ ]. As a rule, all students wear a jersey with their numbers printed on it. So if there are students, each will have a unique jersey number just like a football team. Now, in the beginning, all the students will stand as per the increasing order of their jersey numbers, from left to right. The only difference will be their respective board number which is placed at their respective location. The board location is fixed and cannot be changed.	10
2.		Write a C program to manage a parking lot for the following scenario. You want to keep track of the number of available parking spots, as well as the	

		license plates of the cars parked in each spot. You decide to represent the parking lot as an array of strings, where each string corresponds to a parking spot and contains either the license plate number of the car parked there or an empty string if the spot is available. To keep track of the number of available parking spots, you decide to use a pointer to the first empty spot in the array. Whenever a car leaves the parking lot, you update the array to remove its license plate and increment the pointer to point to the next empty spot. Whenever a car enters the parking lot, you insert its license plate at the current location of the pointer and increment the pointer to point to the next empty spot. Suppose the parking lot is initially empty and you have a pointer p that points to the first element of the array. What happens if you try to insert a car into the parking lot when p is pointing to the last element of the array? How would you handle this situation?	10
3.		Consider this scenario, now you are working as a software developer in a mobile app development company. Your team has been assigned to develop a mobile app for a new startup. The client wants to keep track of the app's features, bugs, and the team working on each task. You have decided to use a C program using structure with an array to manage the app development process. Create a function to read all App Store and write sample output for the following scenario. Example: ===== App Store ===== 1. Add an app 2. View all apps 3. Search for an app 4. Exit Enter your choice: 1 Enter app name: quatumbid Enter app developer :Quad Enter app category :software Enter app price: 25000	10
4.		Create a program to model a banking system. You have a BankAccount class that stores information about each account, including the account holder's name, account number, and balance. You want to add a function to the class that allows account holders to transfer money to another account, but you need to access the private member variables of two BankAccount objects to do so. Write a C++ program that implements a transferMoney() function as a friend function of the BankAccount class. The transferMoney() function should take two BankAccount objects as arguments and transfer a specified amount of money from the first account to the second account. Make sure to include	10

		appropriate member functions in the BankAccount class to store and retrieve account information. Also, make sure to handle cases where the transfer amount exceeds the balance of the first account or where the account numbers are invalid.	
5.		Write a C++ program utilizing constructors and destructors to manipulate an array of non-negative integers, 'arr', by cyclically shifting its elements to the right by k positions. For instance, given an input array arr = [1,2,3,4,5,6,7] and a shift value k = 3, the program should output the transformed array [5,6,7,1,2,3,4]. If shift value k = 1, then the output should be [7,1,2,3,4,5,6]. Illustrate the working of the algorithm and provide comments with explanation for the code. Note: The input array should be passed to the constructor of the "ShiftArray" class from the main function.	10
		Total	**50**

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